



COMPANY PRESENTATION

Associated Engineering Industries

A F P S DIVISION

Automatic Fire Detection & Suppression

Safety & Security Devices for HEMMs

PROTECTING PEOPLE & PROPERTY





AEI

Associated Engineering Industries · AFPS Division

ABOUT US

About Us

Safety through engineering excellence

Associated Engineering Industries (AEI — AFPS Division) specializes in Automatic Fire Protection Systems and allied safety solutions for Heavy Earth Moving Machinery and material-handling equipment used in mining, construction, cement, and industrial applications.

Headquartered in Uppal, Hyderabad, Telangana, AEI has been active in fire protection since 2002 — delivering DGMS-approved systems for off-road and heavy-duty equipment, backed by a strong after-sales service capability across India.

As an independent organisation wholly focused on this domain, AEI supports customers end-to-end — from system selection and supply to installation guidance and long-term service support.

20+

YEARS IN OFF-ROAD FIRE PROTECTION

100%

INDIGENOUSLY DESIGNED & DEVELOPED



Uppal IDA, Hyderabad — Telangana



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Overview

Purpose-built fire-safety for heavy-duty, high-risk operations

2002

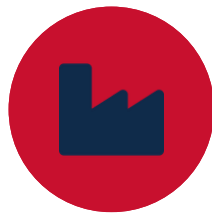
IN THE FIELD SINCE

75%

MINING SECTOR FOCUS

DGMS

APPROVED SYSTEMS



Specialized Manufacturer

Purpose-built fire protection for Heavy Earth Moving Machinery and industrial equipment.



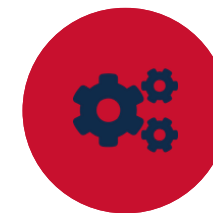
Independent & Focused

An independent organisation fully dedicated to this domain — not a division of a larger conglomerate.



Mining, Construction, Cement

Systems deployed across mining, construction, cement plants and wider industrial applications.



End-to-End Customer Support

From system selection and supply through installation and long-term after-sales service.



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LEADERSHIP

Leadership

Driven by vision, delivered through experience

AEI's growth and operational excellence are the direct result of strong, engineering-led leadership. Our executive team combines deep domain expertise with a commitment to indigenous innovation and customer-first execution.

Under the CEO's direction, AEI has consistently expanded its safety portfolio and earned the trust of India's leading OEMs, mining operators, and industrial customers.

Customer satisfaction is not a metric — it is our paramount objective.



Mr. A V Mallikarjun

CHIEF EXECUTIVE OFFICER

Leads strategy, operations and innovation — championing indigenous engineering and customer-first execution.



Mr. M N Kishore Kumar

MANAGING PARTNER



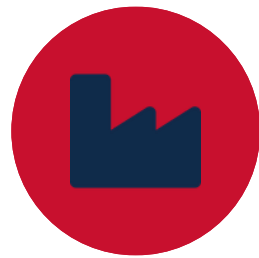
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INDUSTRIES WE SERVE

Industries We Serve

Fire-safety solutions for India's most demanding sectors



OEMs

Original Equipment Manufacturers — factory-fit fire protection systems integrated on production lines.



Mining & Minerals

Public-sector and private operators across coal, lignite and mineral production.



Construction & Cement

Construction sites, cement plants and infrastructure projects operating heavy machinery.



REGULATORY MANDATE

Automatic Fire Protection Systems are mandatory equipment for Heavy Earth Moving Machinery as per the Directorate General of Mines Safety (DGMS).



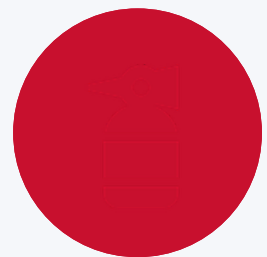
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PRODUCTS & SOLUTIONS

Products & Solutions

A comprehensive safety portfolio beyond fire protection



Fire Protection Systems

Automatic & Semi-Automatic AFPS

- DGMS-approved automatic fire protection
- Configurable 2-Zone and 3-Zone systems
- Engineered per machine type and site needs



Safety & Monitoring

Visibility, awareness, operator safety

- Rear-view camera & digital camera systems
- Audio-visual alarm systems · beacon lights
- Driver fatigue & seat-belt reminder systems



Proximity & Detection

Collision avoidance on active sites

- Radar-based proximity sensors
- Integrated camera + proximity alert systems
- Real-time obstacle detection for operators



Industrial Lighting & Guards

Built for night and low-visibility work

- Heavy-duty LED work lights
- Turbocharger guards to reduce fire risk
- Reliable illumination in harsh conditions



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APPLICATIONS

Applications

Deployed across a wide spectrum of heavy-duty equipment



Excavators

Hydraulic & tracked



Dumpers

Mining haul trucks



Dozers

Crawler & wheel



Tippers

Construction haulage



Wheel Loaders

Quarry & stockyard



Forklifts

Material handling



Reach Trucks

Warehouse & logistics

DEPLOYED ACROSS

Mining · Construction · Cement · Infrastructure · Material Handling · Industrial



AFTER-SALES SERVICE: Trained service engineers deployed at project locations — ensuring continuity of operations across India.



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MAJOR CLIENTS

Major Companies Served by AEI

Trusted by India's leading OEMs, steel producers and mining corporations

OEMs & EQUIPMENT MANUFACTURERS

- ✓ Bharat Earth Movers Ltd. (BEML)
- ✓ TELCON Construction Equipment Co. Ltd.
- ✓ Gainwell Commosales Pvt. Ltd.
- ✓ Sany Heavy Industry India Pvt. Ltd.
- ✓ Scania Automotives

STEEL & MINING CORPORATIONS

- ✓ Steel Authority of India Ltd. (SAIL)
- ✓ National Mineral Development Corp. (NMDC)
- ✓ Western Coalfields Ltd.
- ✓ Northern Coalfields Ltd.
- ✓ Singareni Collieries Co. Ltd.

ALSO TRUSTED BY

TATA · TISCON · Hindustan Zinc · BGR · MCL · XCMG · RITHWIK · Buildcon · AMR · PC Patel · Pro Mining



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SYSTEM INSTALLATION

System Installation Process

Engineered for each machine — installed for long-term reliability

1

Machine Assessment

Size, type and application of the vehicle.

2

Component Placement

Positioned to never obstruct vehicle maintenance or operation.

3

Nozzle & Detection Targeting

Components target identified high-risk fire zones.

4

Hose Routing

Routed to be fully protected from mechanical damage.

5

Accessible Actuators

Manual actuators placed for rapid emergency access.

6

Customer Validation

Final configuration validated with the customer's operations team.



Installed AFPS on mining dumper

Cabinet-mounted cylinders, detection line and actuators integrated into the chassis.



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CERTIFICATIONS & APPROVALS

Standards, Certifications & Approvals

Fully compliant, approved and registered to serve India's most regulated sectors



ISO 9001:2015

Quality Management

Certified QMS governing every step of design, production and service.



ISO 14001:2015

Environmental Management

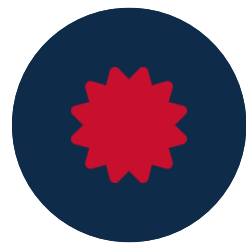
Environmental responsibility built into operations.



DGMS APPROVED

Mines Safety Clearance

DGMS approval — mandatory for fire systems on mining HEMMs.



MSME REGISTERED

Government of India

Registered with the Ministry of Micro, Small and Medium Enterprises.



QMS CERTIFIED

Quality Assurance

Quality Management System certification supporting product reliability.



GHMC TRADE LICENSE

Municipal Compliance

Trade License issued under the GHMC Act — full statutory compliance.



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FIELD INSIGHTS

End-User Challenges & Quick Wins

Real operational hurdles we solve — and fast actions that deliver impact

60%+

OF ALL MINE FIRES

are caused by hot surfaces contacting high-pressure hydraulics and other flammables.

Source: industry field data

COMMON CHALLENGES



Reactive Maintenance

Fix-when-broke cycles instead of preventive care.



Cleanliness & Debris

Oil and dust build-up raising fire risk daily.



Space-Claim Conflicts

Safety kit competing for space with other accessories.

QUICK WINS



Engine / Fan Shutdown

Automatic shutdown and hydraulic pressure relief on detection.



Isolate Hot Surfaces

Physical separation from high-pressure hydraulics and fuel lines.



Operator Training

Empower operators, mechanics and onsite staff on correct system use.



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INNOVATIONS

Our Innovations

Purpose-engineered advances that streamline production and improve site safety



INNOVATIVE ZONAL SYSTEM

2-Zone & 3-Zone Configurations

Flexible zonal protection streamlines production processes, enables targeted suppression and enhances overall system efficiency across different machine sizes.



DUSTFREE CAP TYPE WHEEL

Engineered to keep wheel hubs clean and fully protected from dust and debris during heavy-duty operation.



JUMBO WHEEL CHOCK

Oversized chocks engineered for heavy mining vehicles — providing reliable, stable parking on uneven terrain.



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WHY AEI

There is only one AEI.

A focused specialist — not a division. Not a compromise.

100%

RESOURCE
DEDICATION

of our resources dedicated to off-road
equipment fire protection.

75%

MINING FOCUS

of our business serves the mining
sector — deep domain expertise.

20+

YEARS, ONE MISSION

doing only this, and nothing else, for
over two decades.

1

OF ONE

the only company specializing in
protecting off-road equipment from
fire.

∞

CUSTOMIZABLE

we engineer to customer need, not the
lowest compliance bar.

0

CONGLOMERATE TIES

independent — no internal
competition for resources or attention.



TECHNICAL SCHEMATIC

Reference installation: 3-Zone System on TATA EX-1900 (24 nozzles)





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R & D · LOOKING AHEAD



THE FUTURE OF FIRE SAFETY

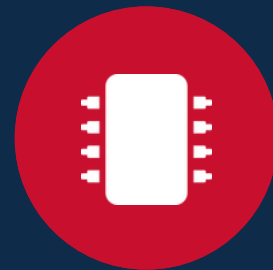
Building the Next-Generation IoT Platform

AEI is actively investing in R&D to evolve our Automatic Fire Protection System into a fully IoT-enabled, cloud-connected safety platform — purpose-built for the next generation of smart, connected heavy machinery.



Real-Time Monitoring

Continuous telemetry from every sensor, cylinder and actuator — live across the entire fleet.



Predictive Diagnostics

Edge intelligence anticipates failures before they happen — turning reactive service into predictive care.



Cloud-Connected Fleet

Unified cloud dashboard giving safety officers a single pane of glass across every site and machine.



Mobile Alerts & Control

Instant smartphone notifications for events, with remote acknowledgment and full audit trails.

A futuristic, connected safety platform — engineered in India, ready for the world.

Thank You.

A Trusted Partner in Fire Safety.

We would welcome the opportunity to discuss how AEI can protect your people, your equipment and your operations.

GET IN TOUCH

Associated Engineering Industries

AFPS Division

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P R O T E C T I N G P E O P L E & P R O P E R T Y